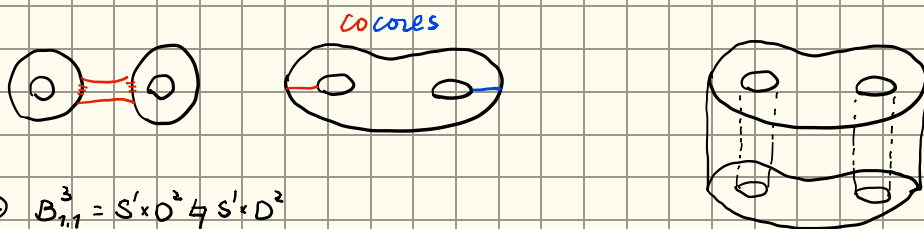


Def: The barbell manifold $B_{i,j}^n := S^i \times D^{n-i} \cup S^j \times D^{n-j}$ ($1 \leq i \leq j \leq n$)

Eg:

① $B_{1,1}^2 = S^1 \times D^1 \cup S^1 \times D^1$ $\times D^1, \times D^2$



② $B_{1,1}^3 = S^1 \times D^2 \cup S^1 \times D^2$



③ $B_{2,2}^4$

Goal:

$$\begin{array}{c} \pi_{i-1}(S^i) \rightarrow \pi_{i-1} \text{Diff}(B_{i,j}^{i+1}) = \pi_{i+j-n} \text{Diff}(B_{i,j}^{i+1}) \xrightarrow{\text{scanning}} \pi_{i+j-n}(\text{Emb}(I, B_{i,j}^{i+1})) \xrightarrow[\substack{\text{intersection} \\ (i+j=n)}]{\text{intersection}} \mathbb{Z} \\ \parallel \\ \mathbb{Z} \\ 1 \end{array} \xrightarrow{\text{Barbell diffeo}}$$

① $B_{i,j}^n = D^{n-j+1} \times B_{i,j}^{j+1}$ (eg. $B_{1,1}^3 = D^1 \times B_{1,1}^2$)

Since $(S^i \times D^{j+1-i} \cup S^j \times D^{n-j}) \times D^{n-j+1} = S^i \times D^{n-i} \cup S^j \times D^{n-j}$ topologically.

Indeed:

$(M_1 \cup M_2) \times D^n = M_1 \times D^n \cup M_2 \times D^n$ (topologically at least)

We regard $M_1 \cup M_2 = M_1 \cup D^{m-1} \times [0,1] \cup M_2$

$h_1: D^{m-1} \hookrightarrow \partial M_1$
 $h_2: D^{m-1} \hookrightarrow \partial M_2$ with some orientation convention.

$M_1 \cup D^{m-1} \times [0,1] \cup M_2 \times D^n = M_1 \times D^n \cup D^{m-1} \times D^n \times [0,1] \cup M_2 \times D^n = M_1 \times D^n \cup M_2 \times D^n$

$(x, y, 0) \sim (h_1(x), y)$
 $(x, y, 1) \sim (h_2(x), y)$

② $D^{n-j+1} \times B_{i,j}^{j+1} \xrightarrow{f} D^{n-j+1} \times B_{i,j}^{j+1}$
 $(x, y) \mapsto (x, f_x)$

$\text{Diff}(B_{i,j}^n) \approx \{ g \in \text{Map}(D^{n-j+1}, \text{Diff}(B_{i,j}^{j+1})) \mid g|_{\partial D^{n-j+1}} = \text{id}_{B_{i,j}^{j+1}} \}$

$= \text{Map}_*(S^{n-j+1}, \text{Diff}(B_{i,j}^{j+1}))$
 $= \Omega^{n-j+1}(\text{Diff}(B_{i,j}^{j+1}))$

$\Rightarrow \pi_{i+j-n} \text{Diff}^F(B_{i,j}^{i+1}) = \pi_{i-1} \text{Diff}(B_{i,j}^{i+1})$



③

I. $B_{i,j}^{j+1} = S^{j+1} \times D^{j+1-i}$ is the once-punctured $S^{j+1} \times D^{j+1-i}$.
 $\underbrace{S^{j+1} \times D^1}_{\text{punctured } D^{j+1}}$
 $M^{j+1} \setminus D^{j+1} = M$

II. Fixing an embedded disk $D^{j+1} \subset \text{Int}(S^{j+1} \times D^{j+1-i}) \subset \underbrace{S^{j+1} \times D^{j+1-i}}_M$

We have a locally trivial fibration of restriction [Palais]

$$\begin{array}{ccc} \text{Diff}(M, \text{rel } \partial M \cup D) & \longrightarrow & \text{Diff}(M, \text{rel } \partial M) \xrightarrow{\sim} \text{Emb}(D^{j+1}, M) \\ \text{ss} & & \downarrow f \longmapsto f|_{D^{j+1}} \\ \text{Diff}(M - \text{Int}(D)) & & \\ \parallel & & \\ \text{Diff}(B_{i,j}^{j+1}) & & \end{array}$$

III. γ is null-homotopic

Indeed, choose a collar $\partial M \times [0, 1] \supset D$.

$$\begin{array}{ccc} \text{We have } \text{Diff}(M, \text{rel } \partial M \times [0, 1]) & \xhookrightarrow{\sim} & \text{Diff}(M, \text{rel } \partial M) \xrightarrow{\sim} \text{Emb}(D, M) \\ & \searrow \text{homotopy equivalence} & \\ & \text{inclusion} & \\ & \Rightarrow \gamma \text{ null homotopic} & \end{array}$$

By long exact sequence, we have

$$\begin{array}{ccccccc} 0 \longrightarrow & \pi_{k+1}(\text{Emb}(D, S^i \times D^{j+1-i})) & \longrightarrow & \pi_k \text{Diff}(B_{i,j}^{j+1}) & \longrightarrow & \pi_k \text{Diff}(S^i \times D^{j+1-i}) & \longrightarrow 0 \\ & \parallel & & & & & \\ & \pi_k(\Omega \text{Emb}(D, S^i \times D^{j+1-i})) & & & & & \end{array}$$

By tubular nbhd

$$\begin{aligned} \text{Emb}(D, S^i \times D^{j+1-i}) &\approx S^i \times D^{j+1-i} \times D_{j+1} \\ &\approx S^i \times D_{j+1} \end{aligned}$$

$$\text{Let } k=i-1: \pi_i(S^i) \hookrightarrow \pi_i \text{Emb}(D, S^i \times D^{j+1-i}) \hookrightarrow \pi_{i-1} \text{Diff}(B_{i,j}^{j+1}) (= \pi_{i,j-1} \text{Diff}^F(B_{i,j}^n))$$

Rmk:

①

$\Omega \text{Emb}(D, S^i \times D^{j+1-i}) \longrightarrow \text{Diff}(B_{i,j}^{j+1})$ is given by iso extension + restriction (Chasing diagrams)

$$\begin{array}{ccc} \text{iso extension } f_+ & & \\ \text{restriction } F_+ & \in & \text{Diff}(B_{i,j}^{j+1}) \\ F_+|_{S^i \times D^{j+1-i} = D} & & \\ \text{s.t. } F_0 = \text{id} & & \\ F_+|_D = 0 & & \end{array}$$

② Isotopy class of $F_+|_{S^i \times D^{j+1-i} = D}$ is independent of the choice of the extension.

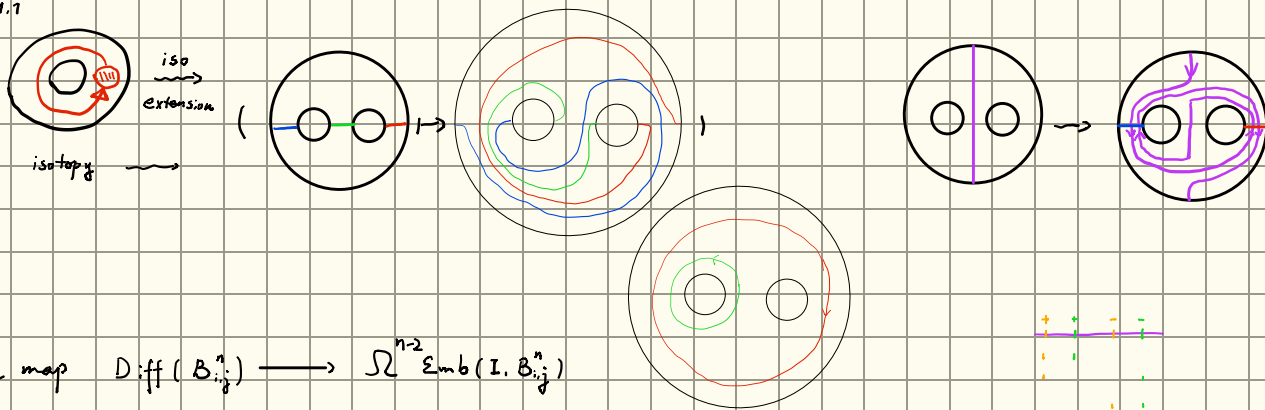
Indeed, if G_+ is another isotopy extension of f
 then $G_+|_D = F_+|_D \Rightarrow F_+ \circ G_+^{-1}$ is an ambient isotopy
 between G_+ and F_+ (Note that $F_+ \circ G_+^{-1}$ fixing $D \forall t$)

$$\begin{array}{ccc} (G_+, \text{const}_0) & \xleftarrow{\text{null homotopic}} & (f, g) \\ \text{ss} \downarrow & & \text{ss} \downarrow \\ (\text{id}_M, g: C_0 \rightarrow C_0) & & \\ \{ (f, g) | g: S^i \times D^{j+1-i} \rightarrow C_0 \} & \xrightarrow{\text{ss}} & \text{Diff}(M) \longrightarrow \text{Emb}(D, M) \\ \uparrow \text{ss} & & \\ \Omega \text{Emb}(D, M) & \longrightarrow & \text{Diff}(M, \text{rel } D) \uparrow \\ \uparrow \text{ss} & & \uparrow \\ \Omega \text{Emb}(D, M) & \longrightarrow & G_+ \end{array}$$

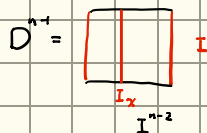
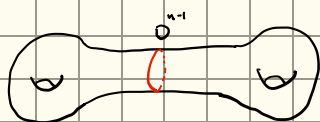
Eg: $B_{1,1}^2$

two Dehn twists

Rule: O_2 might happen.



④ Scanning map $\text{Diff}(B_{i,j}^n) \longrightarrow \Omega^{n-2} \text{Emb}(I, B_{i,j}^n)$



$$\text{Diff}(B_{i,j}^n) \times I^{n-2} \longrightarrow \text{Emb}(I, B_{i,j}^n) \times \text{Emb}(B_{i,j}^n, B_{i,j}^n) \longrightarrow \text{Emb}(I, B_{i,j}^n)$$

$$(f, x) \quad (f|_x, \varphi_x) \quad \varphi_x \circ f|_x$$

φ_x : translation and scaling s.t. $\forall x \in \partial I^{n-1}$ $\varphi_x \circ f|_x =$ a fixed standard embedding of I
+ perturbation

Therefore we have

$$\text{Diff}(B_{i,j}^n) \times S^{n-2} \longrightarrow \text{Emb}(I, B_{i,j}^n)$$

$$\text{Diff}(B_{i,j}^n) \longrightarrow \text{Map}_*(S^{n-2}, \text{Emb}(I, B_{i,j}^n)) = \Omega^{n-2} \text{Emb}(I, B_{i,j}^n)$$

⑤ Intersection number $\Omega^{n-2} \text{Emb}(I, B_{i,j}^n) \longrightarrow \mathbb{Z} \quad (i+j=n)$
 $f \longmapsto \#(f, \text{cores})$

$$f: S^{n-2} \longrightarrow \text{Emb}(I, B_{i,j}^n)$$

$$\tilde{f}: S^{n-2} \times I \longrightarrow B_{i,j}^n$$

$$\{(t_1, t_2) \in I^2 \mid t_1 \leq t_2\}$$

$$F: S^{n-2} \times \Delta_2 \longrightarrow B_{i,j}^n \times B_{i,j}^n$$

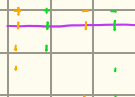
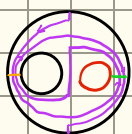
$$(x, t_1, t_2) \longmapsto (\tilde{f}(x, t_1), \tilde{f}(x, t_2))$$

$$\bigcup_{(*) \times D^{n-i}} \times (x \times D^{n-j}) = A \quad \text{codim} = 2n - (n-i + n-j) = i+j = n$$

By perturbation: $F \pitchfork A$

$$\#(F, \text{cores})$$

Eg:



$$\pm 1$$